



GE Medical Systems

Service Manual

Document 2341495

Revision 10

GE Signa® Infinity 1.5T/1.0T Split Head Coil

GE Signa® EXCITE™ 3.0T Split Head Coil

GE 1.5T/ 3.0T HDx Split Head Coil

GE 1.5T HDe Split Head Coil

GE Part Number: 2382458 (3.0T)
2337543 (1.5T)
2337543-2 (1.0T)
5145658 (1.5T HDx/ HDe)
5145659 (3.0T HDx)

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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. • Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. • Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. • Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklinga frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 □ サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 □ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 □□□この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. • 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai anglu valodā. • Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. • Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutra linguagem, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT
(TR)

Bu servis kılavuzunun sadece İngilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu İngilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

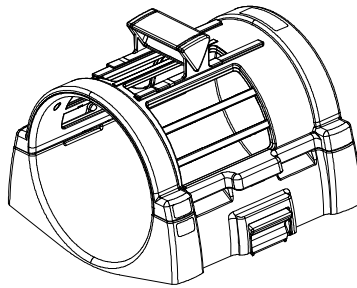
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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE SIGNA 3.0T/1.5T/1.0T Split Head coil, 1.5T/3.0T HDx and 1.5T HDe Split Head Coils



SHIPPING LIST – 3.0T SPLIT HEAD COIL HDx

Description	GE Part#	QTY
3.0T ASSEMBLY HEAD COIL	2376114	1
HEAD COIL QHTR ASSEMBLY 3.0T	5140887	1
SPLIT HEAD COIL FLAT CUSHION	5327298	1
SPLIT HEAD COIL FORMED CUSHION	5327297	1
HEAD RESTRAINT STRAPS	2352202	1
OPERATORS MANUAL	2353925-299	1

SHIPPING LIST – 1.5T SPLIT HEAD COIL HDx/ HDe

Description	GE Part#	QTY
1.5T ASSEMBLY HEAD COIL	2341973	1
HEAD COIL QHTR ASSEMBLY	5140886	1
SPLIT HEAD COIL FORMED CUSHION	5327297	1
SPLIT HEAD COIL FLAT CUSHION	5327298	1
HEAD RESTRAINT STRAPS	2352202	1
OPERATORS MANUAL	2353925-299	1

SHIPPING LIST – 3.0T SPLIT HEAD COIL

Description	GE Part#	Qty
3.0T ASSEMBLY HEAD COIL	2376114	1
QUICK DISCONNECT BOX 3.0T	2376104	1
SPLIT HEAD COIL FLAT CUSHION	5327298	1
SPLIT HEAD COIL FORMED CUSHION	5327297	1
HEAD RESTRAINT STRAPS	2352202	1
OPERATORS MANUAL	2353925-299	1

SHIPPING LIST – 1.5T SPLIT HEAD COIL

Description	GE Part#	Qty
1.5T ASSEMBLY HEAD COIL	2341973	1
QUICK DISCONNECT BOX 1.5T	2336560	1
EXTREMITY ADAPTOR WITH COIL ID	2383622	1
SURFACE COIL ADAPTOR WITH COIL ID	2383623	1
SPLIT HEAD COIL FLAT CUSHION	5327298	1
SPLIT HEAD COIL FORMED CUSHION	5327297	1
HEAD RESTRAINT STRAPS	2352202	1
OPERATORS MANUAL	2353925-299	1

SHIPPING LIST – 1.0T SPLIT HEAD COIL

Description	GE Part#	Qty
1.0T ASSEMBLY HEAD COIL	2341973-2	1
QUICK DISCONNECT BOX 1.0T	2336560-2	1
COIL ADAPTER, MG2 A16 A6 (EXTREMITY)	46-317221G36	1
COIL ADAPTER, MG2 A16 A6 (SURFACE)	46-317222G6	1
SPLIT HEAD COIL FLAT CUSHION	5327298	1
SPLIT HEAD COIL FORMED CUSHION	5327297	1
HEAD RESTRAINT STRAPS	2352202	1
OPERATORS MANUAL	2353925-299	1

1-2 Compatibility

This coil is compatible with the following hardware configurations:

	1.0T Coil	1.5T Coil	3.0T Coil
Signa® EXCITE™ 3.0T			✓
Signa® Infinity 1.5T		✓	
Signa® Infinity 1.0T	✓		
Signa® LX Horizon 1.5T		✓	
Signa® LX Horizon 1.0T	✓		
Signa® TwinSpeed 1.5T		✓	
Signa® Advantage 1.5T		✓	
GE 1.5T HDx		✓	
GE 3.0T HDx			✓
GE 1.5T HDe		✓	

This coil is compatible with the following software configurations:

- 9.1 (M4 and later versions)
- 10.0 and above (10.0 M3 and later)
- 11.0 and above

1-3 Related Documentation

Signa® LX Service Methods CD, 2160623-1

Signa® EXCITE™ 3.0T Service Methods CD, 2389160-1

Signa® Infinity 3.0T/1.5T / 1.0T Split Head Coil Operators Manual, 2353925-299

1-4 Environmental Requirements

Storage/ Shipping Requirements

Category	Value
Temperature	-40 to 70 °C
Humidity	40 to 95 %RH
Atmospheric Pressure	525hPa to 1013 hPa

Dimensions (For reference only)

Description	Part No.	Width (mm)	Length (mm)	Height (mm)
Head Coil 1.0T	2341973-2	420.0	427.0	366.0
Head Coil 1.5T	2341973	420.0	427.0	366.0
Head Coil 3.0T	2376114	420.0	427.0	366.0
QD Box 1.0T	2336560-2	73.0	143.5 (w/o cables)	33.8
QD Box 1.5T	2336560	73.0	143.5 (w/o cables)	33.8
QD Box 3.0T	2676104	81.6	142.2 (w/o cables)	59.0
QHTR 1.5T	5140886	180.4	232.3 (w/o cables)	54.2
QHTR 3.0T	5140887	180.4	232.3 (w/o cables)	54.2

Weight (For reference only)

Description	Part No.	Weight (max)
Head Coil 1.0T	2341973-2	4.5kg
Head Coil 1.5T	2341973	4.5kg
Head Coil 3.0T	2376114	4.5kg
QD Box 1.0T	2336560-2	380g
QD Box 1.5T	2336560	380g
QD Box 3.0T	2676104	360g
QHTR 1.5T	5140886	750g
QHTR 3.0T	5140887	800g

1-5 Theory of Operation

The 1.0T/1.5T/3T/1.5T(HDx)/1.5T(HDe)/3.0THDx assembly head coil is designed to create an evenly distributed magnetic field within its volume. This field is homogeneous inside the coil and rotates in the XY plane at the radio frequency (RF) used to drive the coil. The head coil transmits and receives a RF magnetic field to control the spinning protons in the imaging volume. To do this, it creates an oscillating RF magnetic field equal to the hydrogen precessional frequency. The Larmor equation calculates the precessional frequency by multiplying the gyromagnetic ratio for hydrogen (42.57MHz/Tesla) by the static magnetic field strength, 1.5 Tesla. The 63.86 MHz field produced by the head coil must operate perpendicular (transverse) to the static magnetic field, which is applied along the Z axis of the magnet bore. The same concept applies to the 1.0T and 3.0T head coils with their field frequency at 42.57 MHz and 127.71 MHz respectively.

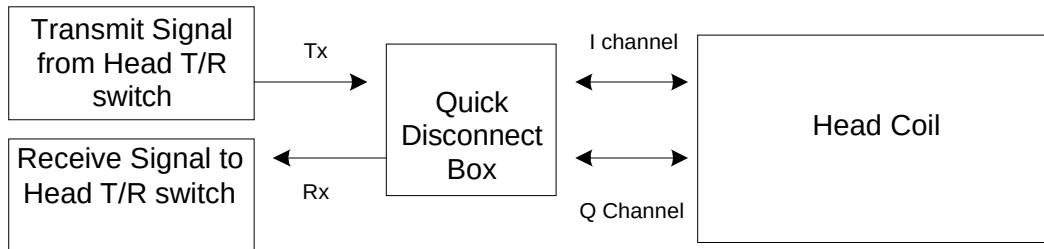
For optimal performance, the following conditions must be met:

- The magnitude of the RF field must be constant for all points in the field of view; the field must meet homogeneous specifications laid out in the SSDS.
- The coil must be tuned to resonate at the Larmor frequency, as described above. The RF head coil is fix tuned and therefore, the resonant frequency is not adjustable past the manufacturing process.

The quick-disconnect box is designed to:

1. Operate as a cable trap to separate the RF ground from the system to the head coil.
2. Act as a power splitter to provide the head coil with 0 and 90-degree power to form a quadrature head coil.

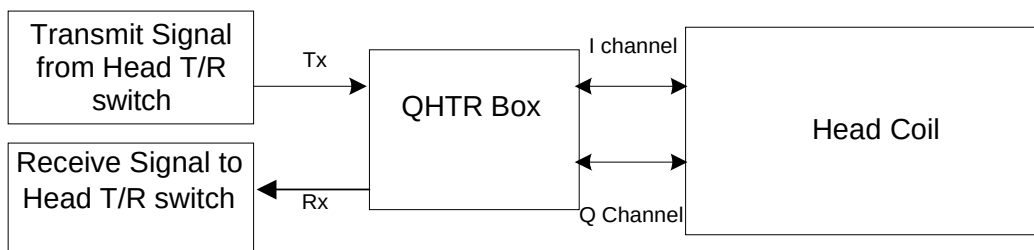
The cable trap will eliminate any standing wave on the ground wire that can be picked up from the environment around head coil. A RF open circuit is created by tuning the ground shield of the cable in the quick-disconnect box to 63.86 MHz at 1.5T.



The QHTR box is applicable to only 1.5/3.0T-HDx and 1.5THDe designed to:

1. Operate as a cable trap to separate the RF ground from the system to the head coil
2. Act as a power splitter to provide the head coil with 0 and 90-degree power to form a quadrature head coil.
3. Acts as T/R(Transmit/Receive) section(applies only to QHTR assembly)

The cable trap will eliminate any standing wave on the ground wire that can be picked up from the environment around head coil. A RF open circuit is created by tuning the ground shield of the cable in the quick-disconnect box to 63.86 MHz at 1.5T.



SECTION 2 - SETUP AND CALIBRATION

2-1 Coil Installation

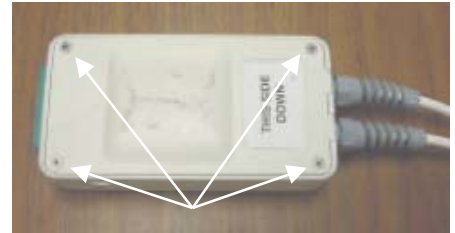
Connect the BNC terminals of the QD box to the coil. The blue labels on the cable and coil should match up, and the cables should not be twisted.

2-1-1 Special Install Notes

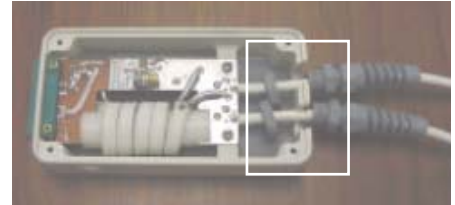
For reverse-ramped magnets, the "I" and "Q" channel cables need to be swapped within the QD box.

For 1.0T and 1.5T coils with the Bendix connector:

- 1) Remove the bottom cover of the QD box by removing the 4 screws marked "A."



- 2) Loosen the strain reliefs on the two BNC cables.



- 3) Swap the two cables



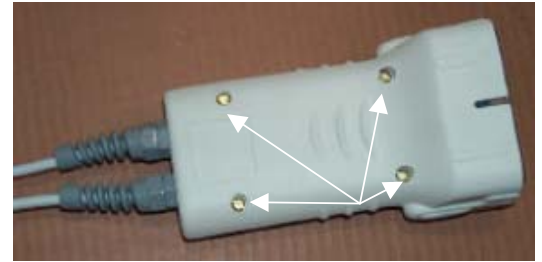
- 4) Retighten the strain reliefs on the two BNC cables

- 5) Replace the bottom cover of the QD box.



For 3.0T coils with the Hypertronics connector:

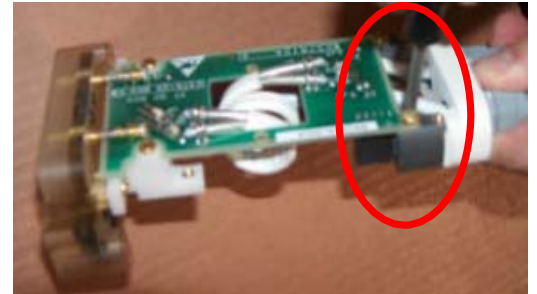
- 1) Remove the top and bottom cover of the QD box by removing the 8 screws, (4 per side).



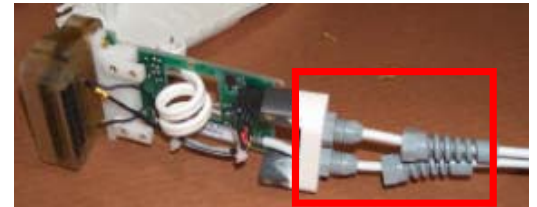
- 2) Remove the 2 screws holding the circuit board to the cable connector.

Caution!

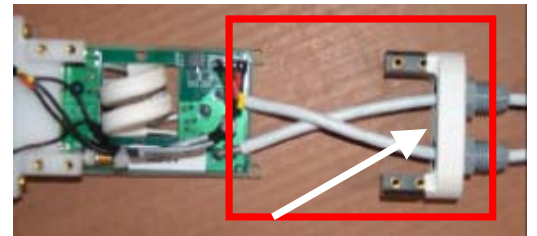
ESD Sensitive Equipment. Make sure to use appropriate precautions before disassembling this equipment.



- 3) Loosen the strain reliefs on the two BNC cables.



- 4) Turn the cable connector upside down, effectively twisting the cables once.



- 5) Attach the circuit board back to the cable connector. Retighten the strain reliefs on the two BNC cables. Replace the top and bottom covers of the QD box.



- 6) On the BNC end of one of the cables, there is a blue label. Cover the blue label with the white rectangular label. Affix a new blue label on the BNC end of the other cable.

**For 1.5T/3T(HDx), 1.5THDe coils
with QHTR**

- 1) Remove the bottom cover of the QHTR by removing the 6 screws.



- 2) Remove the 4 screws holding the circuit board to the connector/cable

Caution!

ESD Sensitive Equipment. Make sure to use appropriate precautions before disassembling this equipment.



- 3) Loosen the strain relief on the two BNC cables.

- 4) swap the two cables.

- 4) Retighten the strain relief on the two BNC cables

- 5) Replace the bottom cover of the QHTR box.



2-1-2 Installing the Coil

The name for this coil is: HEAD

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager..

The cables from the QD box to the head coil should always be straight, with no crossovers or loops evident from the outside.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; System Level Functional Checks; Signal to Noise Check.
2. Perform Section 3 - Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during installation and planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Check plastic screws for tightness, damage, or absence.
3. Perform Section 3-2 - Coil Imaging Performance Verification and record data values in Data Sheet (Table 7-1).
4. If SPT is enabled at the site, use "Head Quick Check" to verify operation.
5. If SNR is trending downwards, check the interconnect pins for damage or abuse.

SECTION 3 - FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; System Level Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TABLE 1 - TOOLS REQUIRED

Field Strength	Description	GE Part #	Qty
1.0T/1.5T	DQA-HEAD SNR PHANTOM	2321556	1
3.0T	TLT Spherical Phantom with Head Loader Shell	2359877 2360031	1 1

3-2-2 Explanation of Procedure

The split head coil can be used in one (1) mode of operation and has one (1) coil name: **HEAD**. Refer to the Data Sheet in Appendix 7-1 to understand the data required to calculate the individual element SNR. All ROI measurements are made on the individual element images, **not** on the composite image.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is an FSE sequence used to minimize susceptibility and B0 inhomogeneity effects. The noise scan is a GRE sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan must be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

3-2-3 Signal Scan

1. From the Scan Desktop, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click [**Patient Position**] to open protocols window.

2. Remove all other surface coils from the cradle. Place the Split Head Coil on the patient table with the BNC connectors extending into the magnet. Lock the baseplate into the table slots. Connect the QD box/QHTR to the BNC connectors on the coil and the connector on the MR unit. The cables from the QD box/QHTR to the head coil should always be straight, with no crossovers or loops evident from the outside.

Remove any pads from the coil. Unlock and remove the top half of the coil by depressing the latches on either side of the coil. Set the phantom into the coil and re-attach the top half of the coil.

3. At the magnet, press “**Alignment Light**” button to turn on the light. Move the cradle to align the coil to the alignment lights. Press “**Landmark**” button to landmark the alignment.

4. Move the coil to scan position by pushing the “**Move to Scan**” button, ensuring cable does not get snagged.

5. At the console, set the protocols per the Signal section from Table 3-2-4: Signal and Noise Protocols. GE Service Personnel can use the Head SNR Signal protocol in the GE / Other protocol folder on 3.0T Systems.

6. Click [**Save Series**] to download the protocols, then click [**Prepare to Scan**].

7. Run [**Auto Prescan**]. Take note of the R1, R2 and TG values.

8. R1 should equal 11, and R2 should equal 14. If not, run [Manual Prescan] and adjust the R1 and R2 values. Click [Done]

9. Run [**Scan**].

3.2.4 Noise Scan

A signal scan must be run **prior** to the noise scan as the same R1, R2 and TG values must be used for both the signal and noise scans. Do **not** run an Auto Prescan prior to the noise scan as the values will be changed.

1. Copy the signal scan series. Use [**Copy Series**] (highlight signal series and click right mouse button) and [**Paste Series**] in **RX Manager**. GE Service Personnel can use the Head SNR Noise protocol in the GE / Other protocol folder on 3.0T Systems.
2. Click [View Edit] and set the protocols per the Noise section from Table 3-2-4: Signal and Noise Protocols.
3. Click [**Save Series**] and click [**Prepare to Scan**].
4. Open [Display CVs] menu under [Research Operations]. Set the "rhformat" and "do_noise" CVs to "1".
5. Run [**Manual Prescan**], do **not** make any changes, and click [**Done**].
6. Run [**Scan**].

TABLE 3-2-4: Signal and Noise Protocols

Protocol	Signal	Noise
Patient Exam Information		
Patient ID	Geservice	geservice
Patient Name	HEAD	HEAD
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)
Patient Position		
Patient Position	Supine	Supine
Patient Entry	Head First	Head First
Coil	HEAD	HEAD
Series Description	Signal	Noise
Imaging Parameters		
Plane	Sagittal	Sagittal
Mode	2D	2D
Pulse Seq	FSE-XL	Gradient Echo
Imaging Options	None	None
PSD Name	Leave Blank	Leave Blank
Protocol	Leave Blank	Leave Blank
Scan Timing		
# of Echoes	1	1
TE	17.0	Min Full
TR	500.0	50.0
Echo Train Length	4	N/A
Bandwidth	31.25	31.25
Additional Parameters		
Flip Angle	N/A	1
Acquisition Timing		
Freq	512	512
Phase	256	256
NEX	1.00	1.00
Phase FOV	1.00	1.00
Freq DIR	A/P	A/P
Auto Center Freq	Peak	Peak
Autoshim	Off	Off
Phase Correct	On	N/A
Contrast	Off	Off
# of Reps B4 Pause	Leave Blank	Leave Blank
Scanning Range		
FOV	24	24
Slice Thickness	3.0	3.0
Spacing	1.5	1.5
Start R/L	0.0	0.0
P/A Center	0.0	0.0
I/S Center	0.0	0.0
End R/L	0.0	0.0
Slices	1	1
Table Delta	0.00	0.00

3.2.5 SNR Image Analysis

SNR Measurement

A rectangular ROI should be used for the signal and noise measurements for 1.0T/1.5T. The actual ROIs should be similar in shape and size to those in Figure 1 below. For 3.0T circular ROI should be used. Select ROI as 80% of the phantom image for signal scan. For Noise scan choose 100% of Rectangular ROI.

ROIs in both signal and noise images can be measured directly in the image browser. Click the user interface button [MEASURE], select the rectangular shape, and adjust its size and orientation so that the ROI is similar to those found in Figure 1. select the circular ROI as 80% of the phantom image. The ROI appear in the lower right corner of the image. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

Record the values for mean signal and mean noise and SNR in Table 4.

Individual receiver SNR is defined as the mean of data within the signal ROI divided by the mean of data within the noise ROI (with a correction factor).

$$\text{SNR} = \frac{\text{Mean of Signal within ROI}}{\text{Mean of Noise within ROI} * 1.253}$$

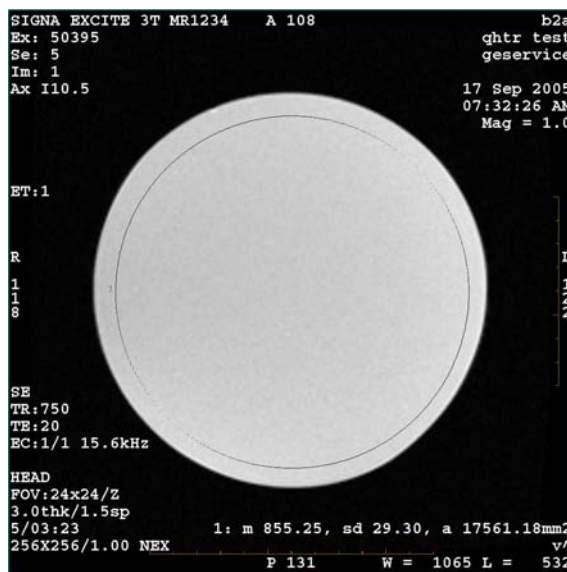
Note: The SNR calculation uses the MEAN of the signal image and MEAN of the noise image



A: Signal ROI Measurement for 1.5T



B: Noise ROI Measurement for 1.5T/3.0T



C: Signal ROI Measurement for 3.0T

Figure 1: Example ROIs

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

TABLE 3- SNR SPECIFICATIONS

Coil	SNR
3.0T Split Head Coil	21.8
1.5T Split Head Coil	29.1
1.0T Split Head Coil	18

3-3 External Cable Check

No cable continuity checking is recommended for the connector because of pin damage concerns.

3-4 PIN Diodes Check

Not Applicable

3-5 Mechanical Hardware Check

Check the BNC cables for mechanical damage and shorted center pins. Inspect the Bendix/ Hypertronics connector for damage from improper insertion. Banana plug on the 1.5T/1.0T Bendix Quick Disconnect plug will damage pins if the operator attempts to insert the connector upside down. At this time also inspect the Bendix/ Hypertronics connector for foreign material and remove as appropriate. Make sure cradle latches and top latches are operating and still have adequate range of motion.

3-6 Troubleshooting Tips

Not Applicable

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution

The coil can be cleaned by wiping with a cloth dampened with 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.

4-2 Special Care Requirements

None

SECTION 5 - REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

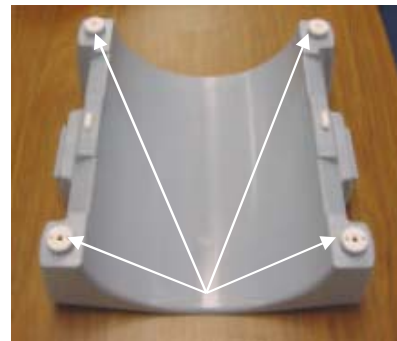
Disconnect old cable with QD box.

Connect new cable with QD box. If coil is reverse ramped, perform section 2-1-1.

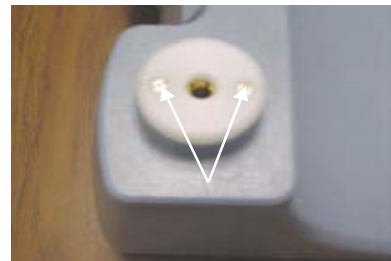
5-2 Mechanical Hardware Replacement

The 4 connection pins on the top half of the coil insert into recessed spring clips on the bottom half of the coil. The spring clips are available as a FRU as referenced in section 6-1.

- 1) The four spring clip connectors are located in the Head coil Base plate, under the plastic caps marked "A"



- 2) Remove the plastic caps by removing the two screws labeled "B"



- 3) Remove the gold spring clip and red gasket.



- 4) Install the new spring clip as shown.



- 5) Insert the red gasket and replace the plastic cap.

5-3 Bottom Cover Plate and Pivot Block Replacement

Refer service instructions 5197695SNT for replacement of Bottom cover plate and pivot blocks supplied as Head coil latching FRU kit 5191687 listed as item J in Sec6-1.

SECTION 6 - RENEWAL PARTS

6-1 Field Replaceable Units

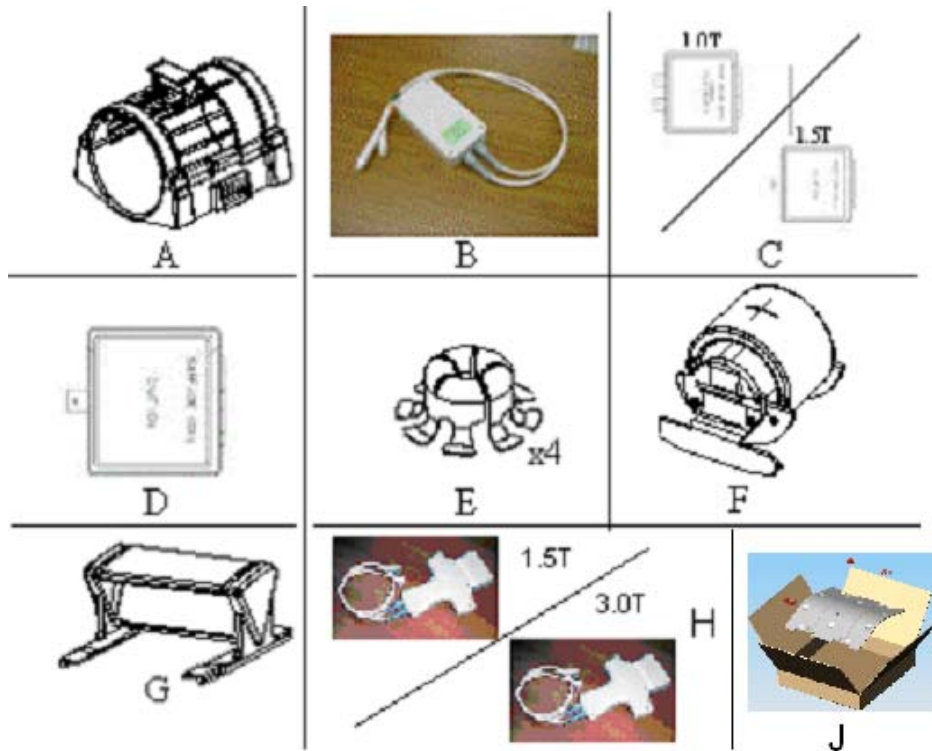


TABLE 2-1: FIELD REPLACEABLE UNITS LIST (3.0T)

Label	Description	GE Part#
A	3.0T ASSEMBLY HEAD COIL (NO CABLE)	2376114
B	QUICK DISCONNECT BOX 3.0T(W/CABLE)	2376104
C	N/A	N/A
D	N/A	N/A
E	SPRING CLIP KIT	2355094
F	TLT SPHERICAL PHANTOM WITH HEAD LOADER SHELL	2359877 2360031
G	MIRROR	2145391
H	QHTR BOX 3.0T (W/CABLE)(FOR HDx)	5140887
J	HEAD COIL LATCHING FRU KIT	5191687

TABLE 2-2: FIELD REPLACEABLE UNITS LIST (1.5T)

Label	Description	GE Part#
A	1.5T ASSEMBLY HEAD COIL	2341973
B	QUICK DISCONNECT BOX 1.5T(W/CABLE)	2336560
C	EXTREMITY ADAPTOR WITH COIL ID	2383622
D	SURFACE COIL ADAPTOR WITH COIL ID	2383623
E	SPRING CLIP KIT	2355094
F	DQA-HEAD SNR PHANTOM	2321556
G	MIRROR	2145391
H	QHTR BOX 1.5T(W/CABLE) (FOR HDx/ HDe)	5140886
J	HEAD COIL LATCHING FRU KIT	5191687

TABLE 2-3: FIELD REPLACEABLE UNITS LIST (1.0T)

Label	Description	GE Part#
A	1.0T ASSEMBLY HEAD COIL (NO CABLE)	2341973-2
B	QUICK DISCONNECT BOX 1.0T (W/ CABLE)	2336560-2
C	COIL ADAPTER, MG2 A16 A6 (EXTREMITY)	46-317221G36
D	COIL ADAPTER, MG2 A16 A6 (SURFACE)	46-317222G6
E	SPRING CLIP KIT	2355094
F	DQA-HEAD SNR PHANTOM	2321556
G	MIRROR	2145391
J	HEAD COIL LATCHING FRU KIT	5191687

TABLE 2-4: OTHER REPLACEABLE ACCESSORIES LIST

Description	GE Part#
SPLIT HEAD COIL CUSHION FORMED	5327297
SPLIT HEAD COIL CUSHION FLAT	5327298
KIT HEAD RESTRAINT STRAPS	2352202

7-1 SNR Data Sheet

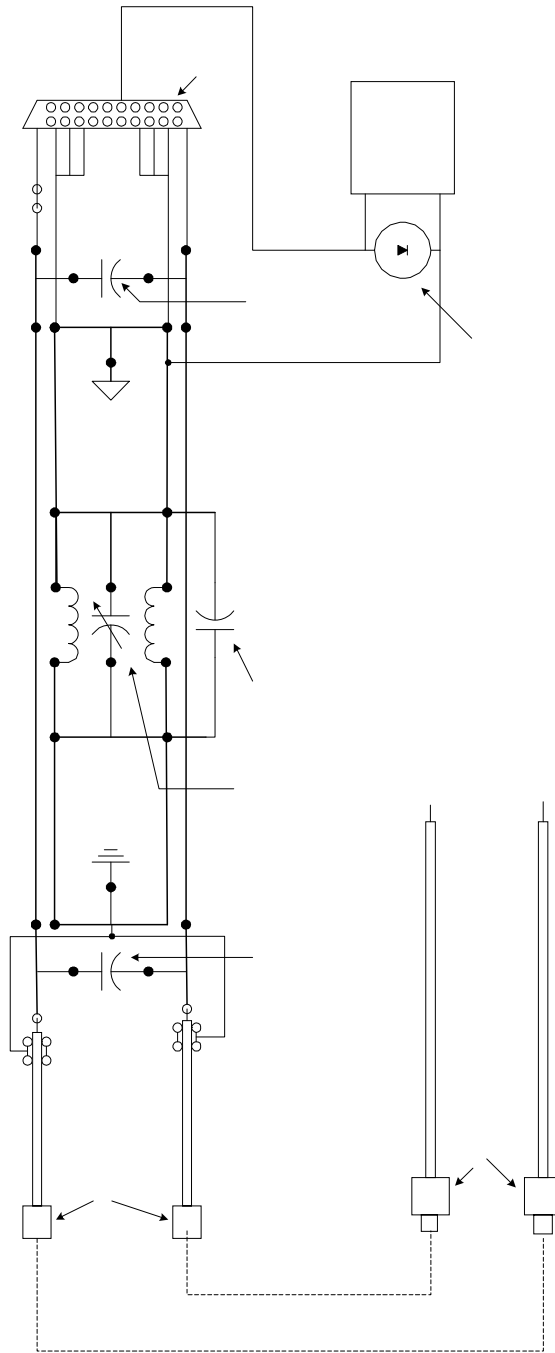
Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Table 7-1: SNR Data Sheet

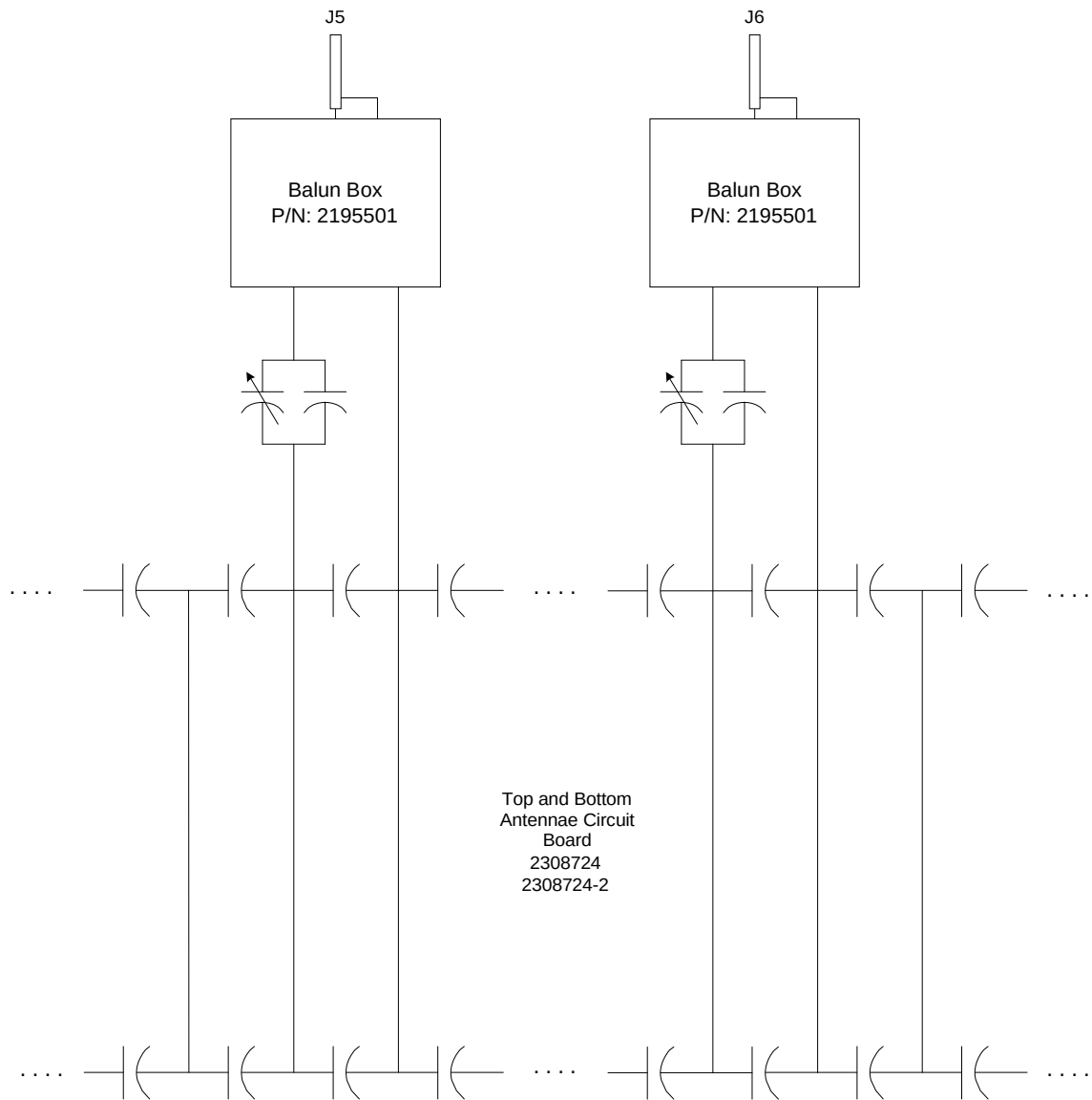
[illegible]

$$SNR = \frac{\text{Mean of Signal within ROI}}{\text{Mean of Noise within ROI} * 1.253}$$

7-2 Quick Disconnect Box Schematic (P/N: 2336560)



From Quick Disconnect Box
P/N: 2336560



NOTE: The schematics for the 3T GP Head Coil is similar to the schematics above except for the part number difference.

7-3 Coil Configuration

Check the system coil configuration for the HEAD coil configuration. If HEAD has not been installed, select HEAD from the list of available coils, and install.

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
0	12/20/2002	K. Zeller	Initial Release
1	12/30/2002	K. Zeller	Title page and Figure 6-1 corrected.
2	1/31/2003	K. Zeller	Table 6-1-1, Table 6-1-2 changed to clarify which FRU includes the cable.
3	3/17/2003	K. Zeller	Updated FRU part numbers
4	5/21/2003	L. Yudkovitch	Included information for 3.0T Head Coil.
5	2/20/2004	K. Hsieh	Changed Op manual to the CD version (2353925-247) Added SNR numbers for section 3-2-5 SNR Image Analysis. Added comment that SNR protocols are only on 3.0T systems. (Section 3-2).
6	12/05/2005	Santhosh	Included HDx/ HDe version with QHTR box
7	12/13/2006	Vijay	Added Sec 5-3 Bottom cover plate and Pivot block replacement (Pg24) Added Head coil latching FRU kit as item J in Sec.6 (Pg 25-26)
8	09/05/2007	Tony	Page 6&7: Shipping list changes to remove Service manual Page 9: Coil & Accessories - Dimensions and Weight updated.
9	15 July 2008	Tony	Humidity and Pressure conditions for storage and shipping changed
10	19 Mar 2009	Ravi	In Shipping list of Head Coils Formed foam and Cushion flat are replaced by new part no (i.e. 2330338 by 5327297 and 2343157 by 5327298).